



WAZUH SIEM DEPLOYMENT AND WINDOWS ENDPOINT MONITORING

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1. Introduction

Security Information and Event Management (SIEM) solutions enable organizations to collect, analyze, and correlate logs from multiple devices in order to detect cyber threats in real time.

Wazuh is a free and open-source SIEM and Extended Detection and Response (XDR) platform that provides centralized security monitoring, log analysis, file integrity monitoring, vulnerability detection, compliance auditing, and endpoint protection.

The purpose of this project was to successfully deploy Wazuh in a virtual environment, connect a Windows endpoint, and demonstrate endpoint monitoring through File Integrity Monitoring (FIM).

2. Task Objectives

The primary objectives of this project were:

- Deploy Ubuntu Server in Oracle VirtualBox.
- Install Wazuh All-in-One.
- Configure networking for communication between the server and Windows host.
- Access the Wazuh Dashboard.
- Deploy a Windows 11 agent.
- Register the Windows endpoint with the Wazuh Manager.
- Configure File Integrity Monitoring.
- Generate security events.

- Verify that alerts are successfully received within the Wazuh Dashboard.
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3. What is Wazuh?

Wazuh is an enterprise-grade SIEM and XDR platform used by Security Operations Centers (SOCs).

Its primary capabilities include:

- Log Collection
- Security Event Correlation
- File Integrity Monitoring (FIM)
- Malware Detection
- Vulnerability Detection
- Security Configuration Assessment (SCA)
- Threat Hunting
- MITRE ATT&CK Mapping
- Endpoint Inventory
- Compliance Monitoring

Wazuh supports Windows, Linux, macOS, and cloud environments.

4. System Requirements

Host Machine

- Operating System: Windows 11
- RAM: 8 GB
- Oracle VirtualBox Installed
- Internet Connection

Virtual Machine

- Ubuntu Server 22.04 LTS
 - RAM: 4 GB
 - CPU: 2 Cores
 - Storage: 25 GB
 - Network Mode: Bridged Adapter
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5. Lab Environment

Component	Description
Host Machine	Windows 11
Hypervisor	Oracle VirtualBox
Server	Ubuntu Server 22.04 LTS
SIEM Platform	Wazuh 4.14.5
Endpoint	Windows 11
Agent	Wazuh Agent 4.14.5

6. Network Configuration

Initially, the virtual machine was configured using NAT networking.

Later, the network adapter was changed to **Bridged Adapter** so that the Windows host and Ubuntu virtual machine could communicate over the same local network.

After configuration, the Ubuntu server received the following IP address:

192.168.18.66

This IP address was later used by the Windows endpoint to connect to the Wazuh Manager.

7. Installing Ubuntu Server

The following steps were performed:

- Created a new virtual machine.
- Allocated 4 GB RAM.
- Assigned 2 CPU cores.
- Attached the Ubuntu Server ISO.
- Installed Ubuntu Server 22.04 LTS.
- Created a local administrator account.
- Logged into the server.
- Verified network connectivity.

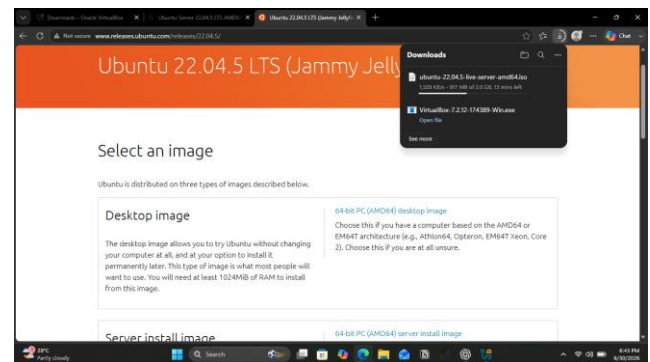


Fig1: Downloading Ubuntu Live Server ISO

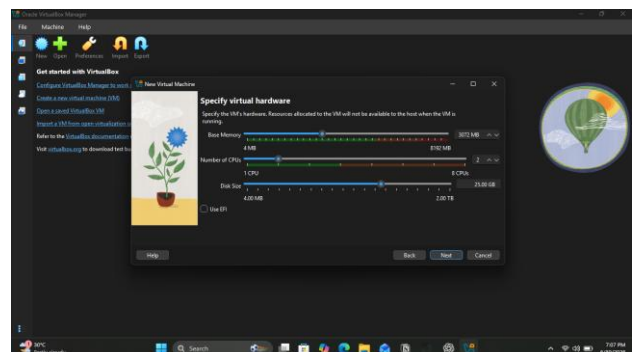


Fig 2: Configuring system hardware for Ubuntu Server VM on VirtualBox

The dashboard provides access to:

- Threat Hunting
- Vulnerability Detection
- File Integrity Monitoring
- Endpoint Inventory
- Security Configuration Assessment
- MITRE ATT&CK Mapping

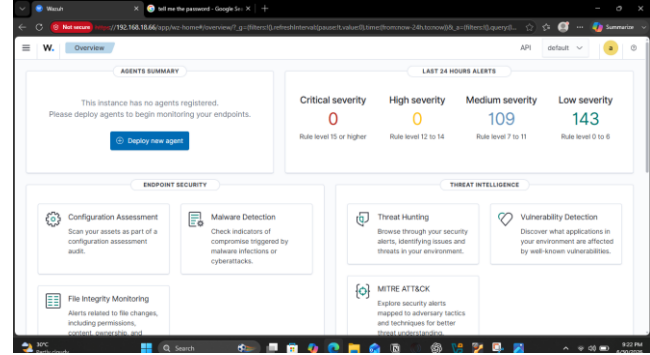


Fig 6: Wazuh Dashboard Accessed after logging in

10. Deploying the Windows Agent

From the dashboard:

Deploy New Agent was selected.

The following options were chosen:

- Operating System: Windows
- Server Address: 192.168.18.66
- Agent Name: Windows11

The dashboard generated a PowerShell installation command

The command downloaded and installed the Wazuh Agent.

Finally, the service was started using:

```
NET START Wazuh
```

The service started successfully.

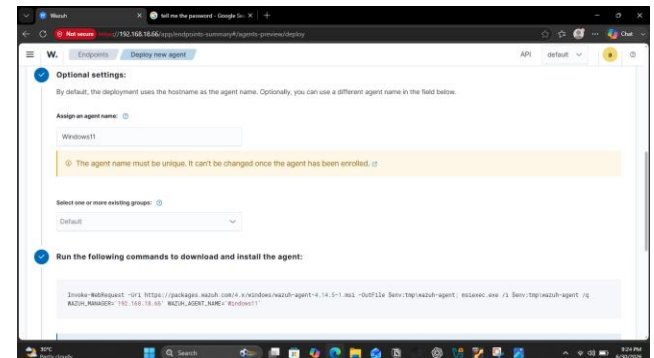


Fig 7: PowerShell Installation command generated

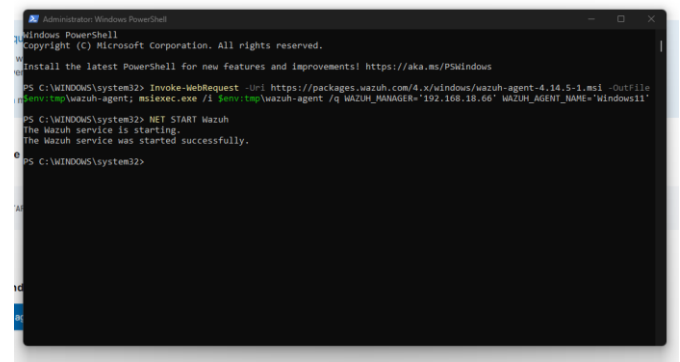


Fig 8: Windows PowerShell

11. Agent Registration

The Windows endpoint successfully registered with the Wazuh Manager.

Within the dashboard, the endpoint appeared as:

- Agent Name: Windows11
- Status: Active
- Operating System: Windows 11 Pro
- Version: 4.14.5

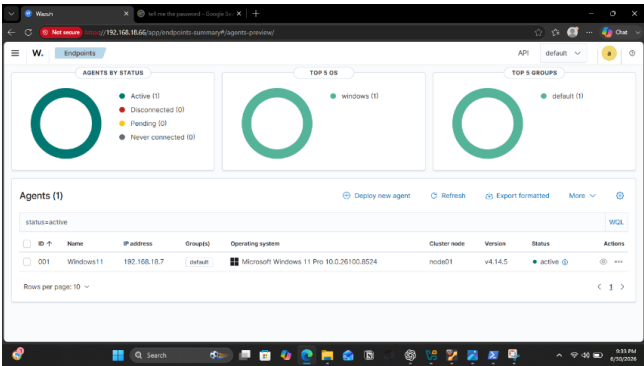


Fig 9: Wazuh Dashboard with Windows Agent Displayed

This confirmed successful communication between the Windows endpoint and the Wazuh Manager.

12. File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) is a security feature that detects changes made to monitored files and directories.

To demonstrate this capability, the Windows agent configuration was modified to monitor the Desktop directory.

The monitored path was:

C:\Users\DELL\Desktop

After updating the configuration, the Wazuh Agent service was restarted.

The manager synchronized the updated configuration successfully.

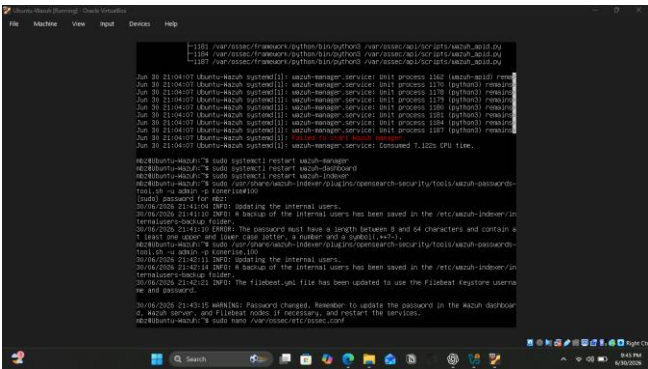


Fig 10: Prompting to edit ossec.conf

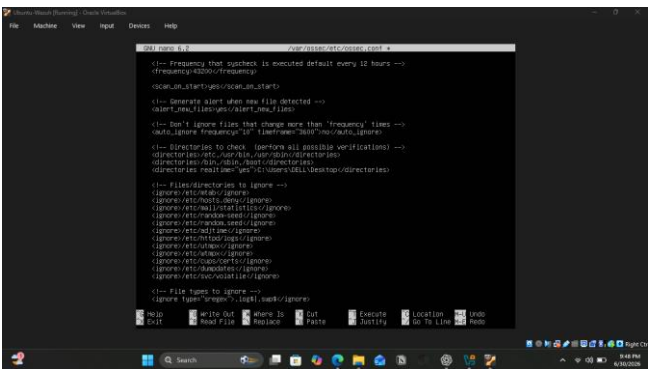


Fig 11: Editing ossec.conf with nano

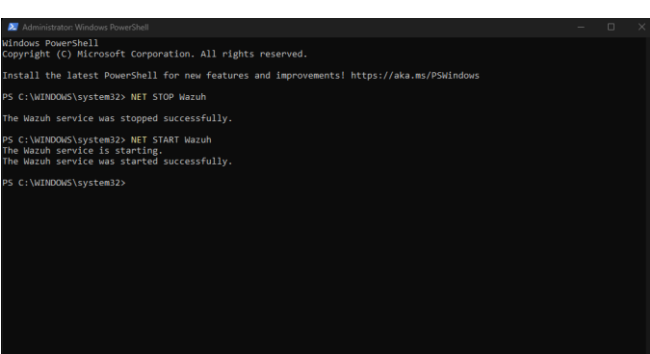
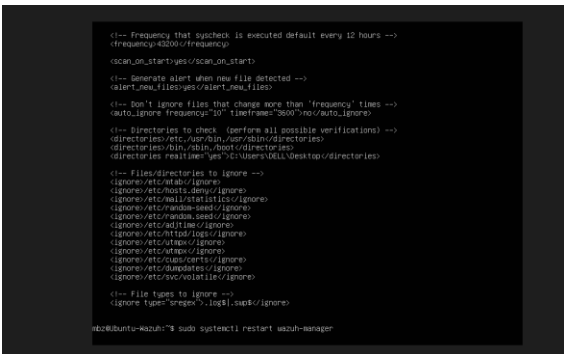


Fig 11 and 12: Restarting Wazuh Services

13. Testing File Integrity Monitoring

To verify File Integrity Monitoring:

1. A test file was created on the Desktop.
2. The file was modified.
3. The file was deleted.

After a short delay, Wazuh generated File Integrity Monitoring alerts.

The alerts appeared within the dashboard and confirmed that:

- File creation was detected.
- File modification was detected.
- File deletion was detected.

This demonstrated successful endpoint monitoring.

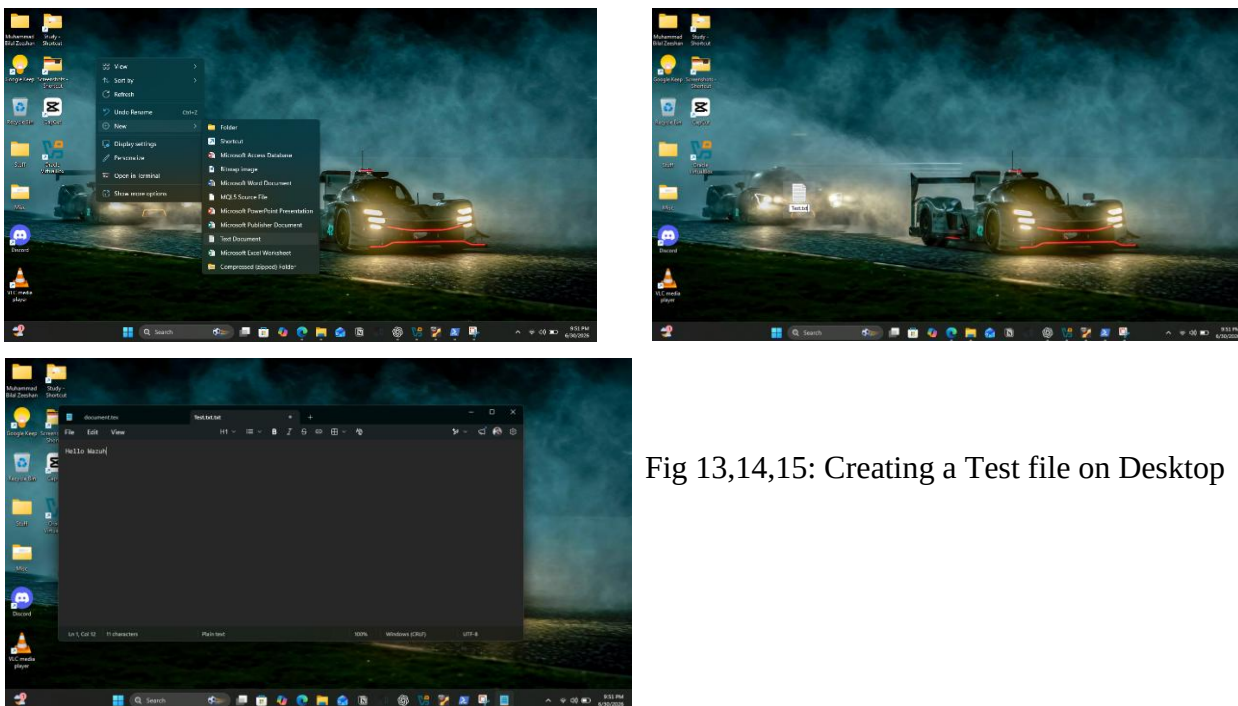


Fig 13,14,15: Creating a Test file on Desktop

A screenshot of the Wazuh File Integrity Monitoring dashboard. The dashboard displays a table of alerts with columns for timestamp, agent name, syscheck path, syscheck event, rule description, and rule level. The table shows a series of alerts for file creation and deletion on the desktop.

timestamp	agent name	syscheck path	syscheck event	rule des...	rule level
Jul 1, 2025 @ 17:32:45.460	Windows11	c:\users\del\Desktop\pff.txt	deleted	File deleted ...	553
Jul 1, 2025 @ 17:31:21.327	Windows11	c:\users\del\Desktop\pff.txt	added	File added ...	554
Jul 1, 2025 @ 17:31:21.268	Windows11	c:\users\del\Desktop\pff\new text document.txt	deleted	File deleted ...	553
Jul 1, 2025 @ 17:31:19.635	Windows11	c:\users\del\Desktop\pff\new text document.txt	added	File added ...	554
Jul 1, 2025 @ 03:49:57.403	Windows11	c:\users\del\Desktop\pff\ghn.txt	deleted	File deleted ...	553
Jul 1, 2025 @ 03:49:55.376	Windows11	c:\users\del\Desktop\pff\ghn.txt	added	File added ...	554
Jul 1, 2025 @ 03:49:55.321	Windows11	c:\users\del\Desktop\pff\ggg.txt	deleted	File deleted ...	553
Jul 1, 2025 @ 03:49:46.736	Windows11	c:\users\del\Desktop\pff\ggg.txt	added	File added ...	554
Jul 1, 2025 @ 03:49:46.728	Windows11	c:\users\del\Desktop\pff\view text document.txt	deleted	File deleted ...	553
Jul 1, 2025 @ 03:49:45.527	Windows11	c:\users\del\Desktop\pff\view text document.txt	added	File added ...	554
Jul 1, 2025 @ 03:33:53.664	Windows11	c:\users\del\Desktop\pff\test.txt	deleted	File deleted ...	553

Fig 16: Alerts received for the creation and deletion of files

14. Troubleshooting

Several issues were encountered during deployment.

Insufficient Memory

The Wazuh installer required at least 4 GB RAM.

Initially, the virtual machine was configured with 3 GB RAM.

The RAM allocation was increased to 4 GB before continuing.

Networking

Initially, the virtual machine used NAT networking.

The network mode was changed to Bridged Adapter so that the Windows endpoint could communicate directly with the Ubuntu server.

Dashboard API Offline

After installation, the dashboard initially displayed API connectivity errors.

Restarting the Wazuh services resolved the issue.

Windows Agent Registration

The Windows endpoint successfully connected to the manager but initially generated no alerts.

Investigation confirmed:

- Agent connectivity
- Active status
- Correct manager IP
- Running services
- File monitoring configuration

After restarting the agent and allowing synchronization to complete, File Integrity Monitoring alerts appeared successfully.

Alert Delay

During testing, alerts appeared after a short delay rather than instantly.

This behavior is expected because Wazuh processes, indexes, and correlates events before displaying them in the dashboard.

15. Results

The project successfully achieved all objectives.

Successfully completed tasks include:

- Ubuntu Server Deployment
 - Wazuh Installation
 - Dashboard Deployment
 - Windows Agent Deployment
 - Endpoint Registration
 - File Integrity Monitoring
 - Security Event Collection
 - Alert Generation
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16. Learning Outcomes

During this project, the following concepts were learned:

- SIEM Architecture
 - Endpoint Monitoring
 - Virtualization
 - Linux Server Administration
 - Wazuh Installation
 - Windows Agent Deployment
 - Security Monitoring
 - File Integrity Monitoring
 - Security Operations Center (SOC) Workflow
 - Troubleshooting Enterprise Security Software
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17. Future Improvements

Future enhancements may include:

- Adding multiple Windows endpoints.
 - Deploying Linux agents.
 - Configuring email alerts.
 - Integrating Sysmon.
 - Implementing Active Response.
 - Enabling VirusTotal integration.
 - Configuring Vulnerability Detection.
 - Integrating Suricata IDS.
 - Monitoring firewall logs.
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18. Conclusion

This project successfully demonstrated the deployment of a complete Wazuh SIEM environment using Ubuntu Server and Oracle VirtualBox.

The Windows endpoint was successfully onboarded using the Wazuh Agent, enabling centralized monitoring and event collection.

File Integrity Monitoring was configured and validated by generating real-time file change alerts.

The project provided practical experience in SIEM deployment, endpoint monitoring, troubleshooting, and security operations. It also demonstrated how Wazuh can be used to enhance an organization's visibility into endpoint activity and improve threat detection capabilities.

Overall, the objectives of the project were successfully achieved, resulting in a fully functional Wazuh SIEM deployment suitable for educational purposes, cybersecurity labs, and SOC training.

19. References

1. Wazuh Documentation – <https://documentation.wazuh.com/>
2. Ubuntu Server Documentation – <https://ubuntu.com/server/docs>
3. Oracle VirtualBox Documentation – <https://www.virtualbox.org/wiki/Documentation>
4. Microsoft Windows Documentation – <https://learn.microsoft.com/>